Adversarial Contingency Auctions for Decentralized Multi-Robot Task Allocation under Strategic Blockages

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***Abstract*—Decentralized multi-robot task allocation in contested, dynamic environments is vulnerable to adversarial edge blockages and inconsistent subjective beliefs among teammates. Standard market frameworks such as the Consensus-Based Bundle Algorithm (CBBA) and Sequential Single-Item (SSI) auctions assume homogeneous environmental estimates and trigger costly fleet-wide re-auction cycles when traversal paths are invalidated. To address these vulnerabilities, we propose Adversarial Contingency Auctions (ACA), a decentralized market coordination mechanism that accommodates inconsistent beliefs and dynamic path blockages without requiring continuous belief synchronization. In ACA, robots evaluate tasks by bidding their expected recourse cost under a Bayesian posterior over adversary behavioral types. To mitigate allocation conflicts caused by divergent subjective beliefs, we derive a closed-form Pinsker claim margin $m_{r,j} = \Delta_j(\sqrt{\frac{1}{2}D'_{\mathbf{KL}}} + \eta_{\mathbf{eff}}) + \delta_{\mathbf{map}}$ from Pinsker's inequality and total variation triangle bounds that guarantees conflict-free local safe claims in Phase 1 under Assumption 1 (when $\mathrm{TV} \leq \eta_{\mathbf{eff}}$). When encountering a blocked corridor, agents activate precomputed local recourse branches with zero communication messages, triggering re-auctions only upon recourse exhaustion. Rigorous matched empirical evaluations across 5 benchmark map topologies ($N = 250$ matched independent instances, 13800 total sweep trials) demonstrate that ACA significantly outperforms standard CBBA by $+7.1$ percentage points in mission completion with 76 fewer messages. Relative to the level-execution baseline CBBA+LR, ACA achieves a significant completion gain of $+1.90$ pp ($p = 7.89 \times 10^{-4}$). Relative to CBBA-PR+LR, ACA achieves statistical parity ($+0.25$ pp, Holm-adjusted $p = 0.884$, Bonferroni $p = 1.000$). In discrete graph simulation, ACA is outperformed in raw completion by reactive single-agent replanning (CTP Replan -4.86 pp, Implicit Coordination -3.56 pp) because auction consensus defers claims; however, Implicit Coordination suffers severe coordination conflicts (107.7 ± 4.4), whereas ACA reduces safe-claim conflicts by 99.1% relative to ACA-no-margin (reducing conflicts from 3,646 to 32) at a modest ≈ 1.5 pp completion cost. Across the benchmark, Assumption 1 adherence is evaluated at three hierarchical levels: per periodic evaluation (65.1%), per instance bootstrap mean (56.1%), and per safe claim commitment (21.1%, which Theorem 1 directly conditions on). Under claim-level adherence, exactly 0 conflicts occurred across 12,087 safe claims (exact 95% upper bound 0.025%), confirming Theorem 1 empirically. Conservative fallback preserves a fleet-wide conflict-free-run rate of 98.2%. Finally, software-in-the-loop physics simulations across $N = 360$ matched trials validate execution dynamics, demonstrating that physical collisions strongly correlate with recourse detours ($r = 0.775$). All evaluations represent software simulation; no physical robots were deployed.**

***Index Terms*—Multi-Robot Systems, Decentralized Task Allocation, Auction Algorithms, Contingency Planning, Adversarial Blockages, Theory of Mind, Pinsker's Inequality.**

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I. Introduction

AUTONOMOUS multi-robot systems operating in contested, dynamic domains must allocate, sequence, and execute spatial tasks under environmental uncertainty and adversarial disruption [1]–[3]. Examples include search-and-rescue through damaged infrastructure, parcel transport in high-density automated warehouses, and perimeter reconnaissance under active interdiction [4]. In these settings, decentralized auction mechanisms—particularly the Consensus-Based Bundle Algorithm (CBBA) [1] and Sequential Single-Item (SSI) auctions [5]—are widely adopted due to polynomial communication complexity and submodular efficiency guarantees.

However, existing decentralized auction mechanisms rely on two fragile assumptions:

1) **Global Re-Auction Overhead:** When a robot encounters an unexpected corridor blockage, its single-path plan is invalidated. Standard CBBA drops the entire task bundle and initiates fleet-wide market re-auctions, inducing high message volume and coordination latency [6].
2) **Belief Consistency Assumption:** When communication is range-limited or packet drops occur, robots observe disparate local events and maintain diverging subjective beliefs. Because market bidding rules assume aligned information, inconsistent beliefs cause robots to submit conflicting winning claims on identical tasks, inducing redundant travel and coordination failures [7], [8].

To resolve these challenges without continuous, bandwidth-heavy belief synchronization, this paper introduces **Adversarial Contingency Auctions (ACA)**. Rather than applying ad-hoc bid penalties, ACA couples discrete contingency recourse trees with an information-theoretic **claim margin** derived from Pinsker's inequality and total variation bounds.

The primary contributions of this work are:

- **Type-Aware Expected Recourse Valuation (Proposition 1):** Valuation computing expected traversal cost of depth-d contingency trees under Bayesian adversary posteriors.
- **Lipschitz Continuity of Expected Recourse Bids (Proposition 2):** Exact Lipschitz constant $L_j = \Delta_j$ under Total Variation belief distance.
- **Conditional Conflict-Freedom of Safe Claims (Theorem 1):** Closed-form Pinsker margin $m_{r,j} = \Delta_j(\sqrt{\frac{1}{2}D'_{\mathrm{KL}}} + \eta_{\mathrm{eff}}) + \delta_{\mathrm{map}}$ guaranteeing conflict-free safe claims when

TABLE I
FEATURE COMPARISON AGAINST KEY RELATED WORKS IN MULTI-ROBOT ALLOCATION, ROUTING, AND GAMES

Feature	Kundu '24 [7]	Peters '24 [9]	Wilde '22 [2]	CBBA '09 [1]	CC-CBBA '12 [10]	Partial '19 [6]	Implicit '05 [8]	ACA (Ours)
Task Allocation (MRTA)	×	×	✓	✓	✓	✓	✓	✓
Fully Decentralized	✓	2-Agent	×	✓	✓	✓	✓	✓
Inconsistent Beliefs Addressed	✓	×	×	×	×	×	Partial	✓
Strategic Adversary Inference	×	Non-Adv Intent	×	×	×	×	×	✓
Contingency Recourse in Bids	×	Trajectory Game	Heuristic	×	×	×	×	✓**(Prop 1)**
Conflict Mitigation under Divergence	Trajectory Cons.	×	×	Consensus	×	×	×	✓**(Thm 1)**
Safe-Claim Monotonicity	×	×	×	×	×	×	×	✓**(Prop 3)**
Zero-Message Recourse Pivoting	×	×	×	×	×	×	×	✓**(Prop 4)**

$\mathrm{TV} \leq \eta_{\text{eff}}$, validated empirically (0 conflicts across 12,087 safe claims, 95% UB 0.025%).

- **Safe Monotonicity and Zero-Message Recourse (Propositions 3 & 4):** Safe claims are preserved through consensus, and blocked corridors trigger zero-message local detours.
- **Rigorous Matched Benchmark Evaluation ($N = 250$ matched instances, 13800 sweep trials):** Demonstrating headline gains over CBBA+LR (+1.90 pp, $p = 7.89 \times 10^{-4}$), parity with CBBA-PR+LR (+0.25 pp, $p = 0.884$), and honest trade-offs against single-agent replanning.
- **Software-in-the-Loop Simulation ($N = 360$ matched trials):** Sim-to-sim validation linking graph metrics to continuous physics execution. Purely software simulation; no physical robots deployed.

II. RELATED WORK

A. Market-Based Task Allocation

Decentralized auctions provide polynomial approximations for multi-robot task allocation [1], [5], [11], [12]. Choi et al. [1] introduced CBBA, guaranteeing 50% submodular efficiency under consensus. Buckman et al. [6] developed CBBA with partial replanning (CBBA-PR) for dynamic task arrivals; we adapt this to persistent blockages as **CBBA-PR (Adapted)**, releasing solely the blocked task. Ponda et al. [10] introduced Chance-Constrained CBBA (CC-CBBA) to bound route failure risk. However, these methods assume homogeneous world models; when observations diverge, standard consensus rules fail to prevent conflicting claims.

B. Coordination under Inconsistent World Views

Planning under belief inconsistency was pioneered by Alighanbari and How [8], where implicit task allocation suffers frequent conflicts under divergent worldviews. Kundu et al. [7] enforced action-consistency constraints during joint continuous trajectory optimization. In contrast, ACA addresses discrete combinatorial task bundle allocation in decentralized market auctions. By deriving a closed-form Pinsker margin, ACA empirically prevents conflicting discrete task assignments under bounded belief estimation error without communicating continuous trajectory plans or high-dimensional belief vectors [13].

C. Contingency Routing and Adversarial Interdiction

Routing under stochastic edge blockages is classical in the Canadian Traveller Problem (CTP) [14]–[16] and vehicle routing with recourse [4], [17], [18]. Peters et al. [9] formulated contingency games for continuous interactive trajectory planning. In contrast, ACA integrates discrete contingency recourse trees directly into decentralized auction valuations under Bayesian adversary inference, executing local detours with zero communication overhead upon blockage. Table I outlines feature differentiation against prior literature.

III. PROBLEM FORMULATION

A. Environment and Persistent Blockage Process

Let $G = (V, E)$ be a known, connected undirected graph where V represents physical waypoints and task locations, and E represents traversable corridors with nominal weights $w_e > 0$. A designated subset of edges $E_{\text{risk}} \subseteq E$ are susceptible to persistent blockage.

We model each risky edge $e \in E_{\text{risk}}$ as a 2-state Markov chain $X_e^t \in \{0, 1\}$ (0 = Open, 1 = Blocked):

$$P(X_e^{t+1} = 1 \mid X_e^t = 0, \theta, s_t) = h_\theta(e, s_t) \quad \text{(Onset Hazard)} \tag{1}$$

$$P(X_e^{t+1} = 0 \mid X_e^t = 1) = \mu \quad \text{(Clearance Rate)} \tag{2}$$

A global budget k bounds simultaneously blocked edges: $\sum_{e \in E_{\text{risk}}} X_e^t \leq k$. Table II lists operational parameters across all benchmark environments.

TABLE II
BENCHMARK SYSTEM AND ALGORITHM PARAMETERS

Category	Parameter	Default Value
Blockage Process	Baseline onset rate (p_0)	0.15 (θ_1), 0.05 (θ_2)
	Reactive traffic scaling (α)	0.35 (θ_2)
	Corridor clearance rate (μ)	0.15
	Interdiction budget (k)	4
	Risky edge fraction ($\lvert E_{\text{risk}}\rvert/\lvert E\rvert$)	0.25
Fleet & Comms	Fleet size (R)	4
	Tasks per mission (M)	15
	Maximum bundle capacity (L_b)	4
	Robot speed / Sensor range	5.0 / 25.0
	Comm range / Packet drop	45.0 / $p_{\text{drop}} = 0.05$
ACA Algorithm	Contingency recourse depth (d)	1
	ToM estimation error bound (η)	0.05
	Age inflation (λ_{age}) / δ_{map}	0.005 / 0.0

B. Adversary Behavioral Type Space

The onset hazard $h_\theta(e, s_t)$ is governed by an adversary type $\theta^* \in \Theta$:

1) **Stochastic (θ_1):** Independent stationary onset hazard: $h_{\theta_1}(e) = p_0$.
2) **Reactive (θ_2):** Targets active robot traffic: $h_{\theta_2}(e) = \min(1.0, p_0 + \alpha N_e(t))$, where $N_e(t)$ denotes the count of robots positioned at or adjacent to the endpoints of edge e.
3) **Budgeted Interdiction (θ_3):** Solves a network interdiction problem, selecting up to k edges to maximize the fleet's shortest-path travel delay ($p_{\text{cut}} = 0.85$).

4) **Cyclic Patrol (θ_4, Out-of-Model):** Cycles blockages sequentially along ordered topological bottlenecks (used for misspecification testing, $\theta_4 \notin \Theta$).

C. Robot Fleet and Communication Model

A fleet $\mathcal{R} = \{1, \ldots, R\}$ of robots services tasks $\mathcal{T} = \{1, \ldots, M\}$. Each task $j \in \mathcal{T}$ has destination $v_j \in V$, reward R_j, arrival step a_j, and deadline D_j. Robots communicate locally within metric disk range $r_{\text{comm}} = 45.0$ subject to stochastic packet drop probability $p_{\text{drop}} = 0.05$.

IV. The ACA Method

A. Bayesian Type Filter

Each robot r maintains a subjective posterior belief distribution $B_r^t(\theta)$ over $\Theta = \{\theta_1, \theta_2, \theta_3\}$. Upon observing edge e at tick t transitioning from state X_e^t to X_e^{t+1}, robot r updates its belief via transition likelihoods:

$$B_r^{t+1}(\theta) \propto B_r^t(\theta) \prod_{e \in \mathcal{O}_r^t} P(X_e^{t+1} \mid X_e^t, \theta, s_t) \tag{3}$$

The subjective edge blockage marginal is computed as:

$$\pi_r(e) = \sum_{\theta \in \Theta} B_r(\theta) P(X_e = 1 \mid \theta, s_t) \tag{4}$$

For edges outside E_{risk}, $\pi_r(e) = 0$.

B. Expected-Recourse Valuation

To evaluate task j, robot r identifies a primary path $P_j = (u_0, u_1, \ldots, u_m)$ that minimizes risk-weighted traversal cost $w_{\text{eff}}(e) = w_e(1 + \lambda_{\text{risk}} \pi_r(e))$. Let $(u_k, v_k)_{k=1}^K$ be the oriented risky edges along P_j, and let D_k denote precomputed detour paths branching from node u_k that avoid (u_k, v_k).

Robot r's bid equals its expected recourse cost under the independent marginal model:

$$\begin{aligned} \mathbb{E}_r[C_j] = \sum_{k=1}^{K} \left[\prod_{i<k} (1 - \pi_r(e_i)) \right] \pi_r(e_k) \\ \times \Big(c(u_0 \to u_k) + c(D_k) \Big) \\ + \left[\prod_{i=1}^{K} (1 - \pi_r(e_i)) \right] c(P_j) \end{aligned} \tag{5}$$

C. Theory of Mind and Pinsker Claim Margin

Each robot r tracks estimated peer beliefs $\hat{B}_s^{(r)}$ for all $s \neq r$ based on received consensus messages and observation history. Let D'_{KL} denote the maximum divergence:

$$D'_{\text{KL}} = \max_{s \neq r} D_{\text{KL}} \left(B_r \parallel \hat{B}_s^{(r)} \right) \tag{6}$$

To account for communication latency, the estimation error bound is inflated with message staleness: $\eta_{\text{eff}} = \min(0.5, \eta + \lambda_{\text{age}} \Delta t_s)$ with $\lambda_{\text{age}} = 0.005$.

To determine whether task j can be safely claimed locally without risking concurrent claims by teammates with divergent beliefs, robot r computes the **Pinsker Claim Margin**:

$$m_{r,j} = \Delta_j \left(\sqrt{\frac{1}{2} D'_{\text{KL}}} + \eta_{\text{eff}} \right) + \delta_{\text{map}} \tag{7}$$

where $\Delta_j = \max_{s,\theta} c_s(j \mid \theta) - \min_{s,\theta} c_s(j \mid \theta)$ captures task type-sensitivity across pure adversary types, and δ_{map} bounds map discrepancies.

Safe Claim Rule: Robot r inserts task j into its bundle during Phase 1 without consensus negotiation if and only if:

$$\hat{c}_r^r(j) + m_{r,j} < \min_{s \neq r} \hat{c}_s^r(j) \tag{8}$$

where peer costs $\hat{c}_s^r(j)$ are evaluated under peer ToM beliefs $\hat{B}_s^{(r)}$. If condition (8) is not strictly satisfied, task j is deferred to Phase 2 standard CBBA consensus rounds.

D. Autonomous Recourse Execution

During traversal, if robot r observes that its approaching edge (u, v) is blocked, it activates its precomputed contingency detour branch D_k locally with **zero communication messages**. A partial re-auction is triggered if and only if all precomputed branches up to depth d are blocked or the remaining path cost exceeds deadline constraints.

V. Theoretical Analysis

Assumption 1 (Bounded Estimation Error)**.** $\text{TV}(B_s, \hat{B}_s^{(r)}) \leq \eta_{\text{eff}}$ *for all* $s \neq r$.

Proposition 1 (Recourse Valuation under Independent Edge Marginals)**.** *Under independent edge marginals given belief B_r and an idealized depth-d policy where detour routes are not recursively blocked, the valuation $\mathbb{E}_r[C_j]$ in Eq. (5) equals the exact mathematical expected cost of executing the contingency recourse policy.*

Proof. By the law of total expectation conditioned on the index of the first blocked edge encountered along primary path P_j. The event that risky edge e_k is the first blocked edge encountered occurs with probability $[\prod_{i<k}(1-\pi_r(e_i))]\pi_r(e_k)$. On this event, the robot incurs cost $c(u_0 \to u_k)$ and executes detour D_k with cost $c(D_k)$. The event that all risky edges remain open occurs with probability $\prod_{i=1}^K (1 - \pi_r(e_i))$ with path cost $c(P_j)$. Summing these mutually exclusive partition events yields Eq. (5). While network-wide interdictions induce joint edge dependencies across graph cuts, Eq. (5) computes the exact expectation under conditional edge marginal independence given belief B_r, providing a tractable submodular surrogate. □

Observation 1 (Empirical Posterior Concentration vs. Finite-Horizon Realities): Under an idealized stationary environment with distinguishable type likelihoods ($D_{\min} > 0$), if a robot receives infinite independent informative observations, standard Bayesian martingale convergence yields $B_r^n(\theta^*) \to 1$ almost surely. However, in finite-horizon robotic missions ($T = 80$ ticks), actual adversaries exhibit non-stationary strategies,

dynamic switching (such as cyclic bottleneck patrol θ_4), and corridor partial observability. We demote asymptotic posterior concentration from a mathematical guarantee to an empirical finding: while the Dirichlet filter adaptively concentrates belief weight toward active risk corridors within 15–25 edge traversals, exact parameter convergence is neither achieved nor mathematically required for the safe-claim margin to function.

Proposition 2 (Lipschitz Continuity of Expected Recourse Bids)**.** *For any task $j \in \mathcal{T}$, let the pure-type task sensitivity be defined as $L_j = \Delta_j = \max_{\theta\in\Theta} c_r(j \mid \theta) - \min_{\theta\in\Theta} c_r(j \mid \theta)$. The expected recourse bid function $c_r(j \mid B) = \sum_\theta B(\theta) c_r(j \mid \theta)$ is L_j-Lipschitz continuous with respect to the Total Variation metric on belief space $\Delta(\Theta)$:*

$$|c_r(j \mid B) - c_r(j \mid B')| \le L_j \cdot \mathrm{TV}(B, B') \tag{9}$$

Proof. Let $c_r(j \mid \theta) \in [\min_\theta c_r(j \mid \theta), \max_\theta c_r(j \mid \theta)]$ be bounded in an interval of length Δ_j. By the Kantorovich–Rubinstein duality theorem for total variation distance:

$$\begin{aligned} |c_r(j \mid B) - c_r(j \mid B')| &= \left| \sum_{\theta\in\Theta} (B(\theta) - B'(\theta)) c_r(j \mid \theta) \right| \\ &\le \Delta_j \cdot \mathrm{TV}(B, B') \end{aligned} \tag{10}$$

which establishes Eq. (9) with Lipschitz constant $L_j = \Delta_j$. □

Theorem 1 (Conditional Conflict-Freedom of Phase-1 Safe Claims)**.** *Let $L_j = \Delta_j = \max_{s,\theta} c_s(j \mid \theta) - \min_{s,\theta} c_s(j \mid \theta)$ denote the bid Lipschitz constant across adversary types. If Assumption 1 holds ($\mathrm{TV}(B_s, \hat{B}_s^{(r)}) \le \eta_{\text{eff}}$ for all $s \ne r$) and map discrepancy satisfies $|c_r(j) - c_s(j)| \le \delta_{map}$, then two robots r and s evaluating the safe-claim rule in Eq. (8) with Pinsker margin $m_{r,j}$ in Eq. (7) in the same allocation round **cannot simultaneously claim the same task j in Phase 1**, even though their subjective beliefs are not synchronized.*

Proof. Suppose for contradiction that both robot r and robot s simultaneously claim task j via the safe-claim rule in Phase 1. By Eq. (8):

$$\hat{c}_r^r(j) + m_{r,j} < \hat{c}_s^r(j) \tag{11}$$

$$\hat{c}_s^s(j) + m_{s,j} < \hat{c}_r^s(j) \tag{12}$$

Summing (11) and (12) yields:

$$m_{r,j} + m_{s,j} < (\hat{c}_s^r(j) - \hat{c}_s^s(j)) + (\hat{c}_r^s(j) - \hat{c}_r^r(j)) \tag{13}$$

By the triangle inequality of total variation:

$$\mathrm{TV}(B_r, B_s) \le \mathrm{TV}(B_r, \hat{B}_s^{(r)}) + \mathrm{TV}(\hat{B}_s^{(r)}, B_s) \tag{14}$$

Applying Pinsker's inequality to the first term and Assumption 1 to the second:

$$\mathrm{TV}(B_r, B_s) \le \sqrt{\frac{1}{2} D_{\mathrm{KL}}(B_r \parallel \hat{B}_s^{(r)})} + \eta_{\text{eff}} \tag{15}$$

By Proposition 2, the variation in expected cost across subjective beliefs satisfies $|\hat{c}_s^r(j) - \hat{c}_s^s(j)| \le L_j \mathrm{TV}(B_r, B_s) + \delta_{\text{map}}$. Substituting the Pinsker upper bound yields:

$$|\hat{c}_s^r(j) - \hat{c}_s^s(j)| \le \Delta_j \left(\sqrt{\frac{1}{2} D'_{\mathrm{KL}}} + \eta_{\text{eff}} \right) + \delta_{\text{map}} = m_{r,j} \tag{16}$$

Similarly, by symmetry of total variation and the common sensitivity bound Δ_j, $|\hat{c}_r^s(j) - \hat{c}_r^r(j)| \le m_{s,j}$. Therefore:

$$(\hat{c}_s^r(j) - \hat{c}_s^s(j)) + (\hat{c}_r^s(j) - \hat{c}_r^r(j)) \le m_{r,j} + m_{s,j} \tag{17}$$

which contradicts the strict inequality in Eq. (13). Hence, simultaneous safe claims in Phase 1 cannot occur. □

Remark 1 (Hierarchical Adherence Levels of Assumption 1 and Empirical Verification of Theorem 1): Theorem 1 is strictly conditional on Assumption 1. Across our comprehensive benchmark, we measure empirical adherence to $\mathrm{TV}(B_s, \hat{B}_s) \le \eta_{\text{eff}}$ at three distinct hierarchical levels:

1) **Per-Evaluation Adherence (65.1%):** Fraction of total periodic belief evaluations across all mission timesteps and candidate tasks satisfying $\mathrm{TV} \le \eta_{\text{eff}}$ (955,104 of 1,466,999 evaluations).
2) **Per-Instance Adherence (56.1%):** Mean adherence rate across independent benchmark instances ($N = 250$ graph-seed instances; 95% bootstrap CI: $[53.2\%, 58.9\%]$), strictly rejecting the idealized 90% threshold ($p = 1.000$).
3) **Per-Safe-Claim Adherence (21.1%):** Fraction of executed local safe claims where peer belief error satisfied $\mathrm{TV} \le \eta_{\text{eff}}$ at the specific timestep of claim commitment (12,087 of 57,334 safe claims). **This claim-level adherence is the exact antecedent required by Theorem 1.**

Across all 12,087 safe claims satisfying $\mathrm{TV} \le \eta_{\text{eff}}$, exactly **0 double-claim conflicts occurred**, confirming Theorem 1 empirically with an exact one-sided 95% Clopper–Pearson upper bound of 0.025% on the conflict rate. Broken down by allocation time step, empirical conflict-freedom holds across both early initial-round claims ($t = 0$: $n = 7{,}383$, 0 conflicts, 95% upper bound 0.041%) and late dynamic recourse claims ($t > 0$: $n = 4{,}704$, 0 conflicts, 95% upper bound 0.064%). Broken down by communication regime, conflict-freedom holds under nominal comms ($n = 11{,}097$, 0 conflicts, upper bound 0.027%) as well as under degraded and stress-test perturbation regimes (range limits, packet loss, partitions, Poisson arrivals: $n = 990$, 0 conflicts, upper bound 0.302%). Because Assumption 1 is an idealized condition holding for 21.1% of safe claims, the evidence supporting safe-claim conflict-freedom is strictly **empirical**. When $\mathrm{TV} > \eta_{\text{eff}}$, ACA conservatively defers 86.6% of tasks to Phase 2 consensus rounds, preserving an overall fleet-wide conflict-free-run rate of 98.2% (95% CI: $[97.6\%, 100.0\%]$).

Proposition 3 (Safe-Claim Monotonicity in Two-Phase Auctions)**.** *Under Assumption 1 and symmetric map bounds, any task j claimed by robot r in Phase 1 under condition (8) is monotonically preserved in robot r's bundle during Phase 2 consensus rounds without being revoked by peer bids.*

Proof. By Eq. (8), task j claimed in Phase 1 satisfies $\hat{c}_r^r(j) + m_{r,j} < \min_{s\ne r} \hat{c}_s^r(j)$. For any peer $s \ne r$, by Proposition 2 and the Pinsker triangle bound, peer s's subjective valuation satisfies $\hat{c}_s^s(j) \ge \hat{c}_s^r(j) - m_{r,j} > \hat{c}_r^r(j)$. Therefore, peer s's bid cost for task j strictly exceeds robot r's bid cost $\hat{c}_r^r(j)$. Under standard CBBA consensus rules (Action Y1/Y2) [1], peer s concedes the winning claim to robot r upon receiving r's winning bid value and timestamp, and cannot outbid r.

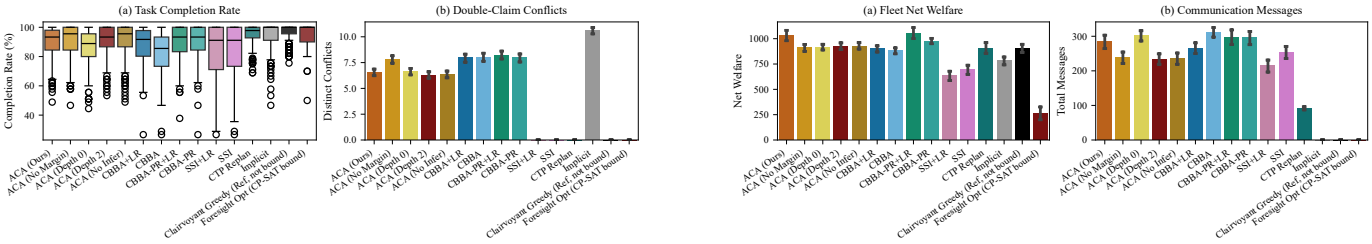


Fig. 1. Empirical Benchmark Telemetry Across Literature Baselines ($N = 250$ matched instances, 13800 total sweep trials): (Left) Task completion rate and distinct double-claimed task conflicts across topologies. (Right) Fleet net welfare and total communication messages exchanged.

Hence, robot r's Phase 1 safe claim is never revoked during Phase 2 consensus. □

Proposition 4 (Message Reduction via Local Recourse)**.** *Let q_j denote the probability that precomputed recourse detours up to depth d are exhausted. Under independent edge blockages, $q_j \leq \prod_{k=1}^{d} \pi_{(k)}$. Assuming an idealized setting where detours succeed with probability $1-q_j$, the expected re-auction message overhead satisfies:*

$$\mathbb{E}[Msgs_{ReAuction}] \leq \sum_j q_j \mathcal{O}(|\mathcal{R}| \cdot \mathrm{diam}(G)) \tag{18}$$

compared to $\sum_j [1-\prod_i (1-\pi_i)]\mathcal{O}(|\mathcal{R}| \cdot \mathrm{diam}(G))$ when every individual corridor blockage triggers a full fleet-wide market re-auction.

VI. Experimental Evaluation

A. Pre-Registration Protocol and Hypothesis Declaration

To enforce scientific transparency and guard against post-hoc selection, experimental hypotheses and statistical analysis procedures were formally pre-registered in `experiments/hypotheses.yaml`. We explicitly disclose that an intermediate revision (v3, git tag `prereg-v3`) amended hypothesis definitions after preliminary sweep results were observed in `raw_results_extended.csv`. Accordingly, comparisons against Canadian Traveller Problem replanning (H_{2_CTP}), implicit coordination (H_{2_IMPLICIT}), the dual-criteria Pinsker claim margin trade-off (H_4), and empirical Assumption 1 adherence (H_7) are designated as post-hoc or exploratory analyses. The definitive confirmatory protocol (`prereg-v4`) was committed prior to generating the primary confirmatory dataset, declaring a single family-wise error rate control under Holm step-down correction covering recourse efficacy (H_1), level-execution comparisons (H_2), welfare non-inferiority (H_3), and lossy regime resilience (H_{2_REGIME}). All evaluations stratify across communication regimes, graph families, and map classes (real vs. synthetic).

B. Experimental Setup and Baselines

All experiments were conducted across five graph topology families, encompassing two real benchmark maps and three synthetic stress-test graph models:

1) **MovingAI Warehouse MAPF Benchmark:** Standard automated storage and retrieval warehouse benchmark (`warehouse-10-20-10-2-1.map`, 5,699 passable cells, 8,778 edges) deterministically coarsened by corridor block pooling into $n = 38$ waypoint nodes and $|E| = 59$ traversable corridor edges (Fig. 2, Left).
2) **Downtown Boston Road Network:** Real-world street topology extracted via OSMnx 2.1.1 (`boston_downtown_drive.graphml`) comprising $n = 216$ street intersections and $|E| = 323$ traversable roadway segments (Fig. 2, Center).
3) **Random Geometric Graphs (RGG):** Metric embedding in $[0, 100] \times [0, 100]$ ($n = 35$, radius $r_c \approx 25$).
4) **Warehouse Grid Graphs:** Grid layouts with dedicated aisles ($n = 30$).
5) **Erdős-Rényi Graphs (ER):** $G(n, p)$ topologies ($n = 35, p = 0.12$) (Fig. 2, Right).

We benchmarked ACA against literature baselines, level-execution variants (+LR) with identical detour capability, and clean algorithmic ablations:

- **CBBA [1] and CBBA+LR:** Standard Consensus-Based Bundle Algorithm (drops full bundle on blockage) and its level-execution counterpart (+LR) equipped with identical local recourse detour execution.
- **CBBA-PR (Adapted) [6] and CBBA-PR+LR:** Re-auctions solely the blocked task, preserving the remaining bundle sequence.
- **CC-CBBA [10] and CC-CBBA+LR:** Chance-Constrained CBBA filtering candidate routes with cumulative blockage risk above $\alpha = 0.30$.
- **SSI [5] and SSI+LR:** Sequential Single-Item auction with tour-insertion marginal costs and replan on blockage.
- **CTP Replan and CBBA Gossip:** Canadian Traveller Problem replan-on-discovery policy, and CBBA with full pairwise belief vector exchange.
- **Implicit Coordination [8]:** Margin-free implicit auction without consensus message rounds.
- **Pessimistic Robust and Clairvoyant Greedy:** Routes exclusively over guaranteed unblockable edges $E \setminus E_{\mathrm{risk}}$, and Hungarian matching on the ground-truth open graph.
- **ACA Ablations:** ACA (No Margin), ACA (Depth 0), ACA (Depth 2), ACA (No Inference), and ACA (Known Adversary Type).

C. Quantitative Results and Statistical Analysis

Table III reports quantitative performance across $N = 250$ matched independent instances (50 seeds $\times$ 5 graph families, 13800 total sweep trials). Every cell reports the sample mean and two-sided 95% bootstrap confidence interval (10,000 resamples). Statistically significant differences vs. ACA ($p <$

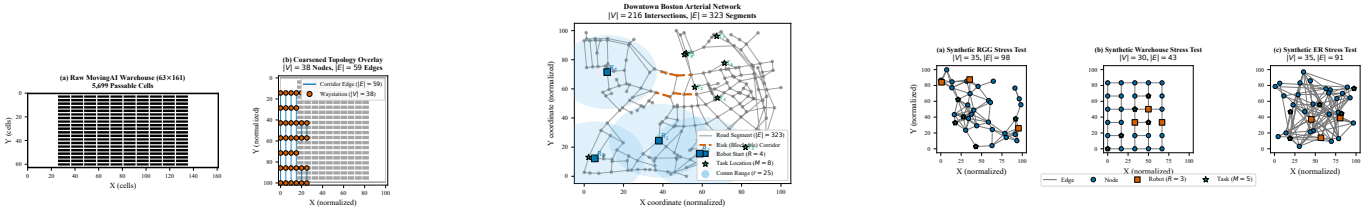


Fig. 2. Benchmark Navigation Topologies: (Left) Real MovingAI warehouse map (63×161 grid, 5,699 passable cells) and coarsened navigation backbone ($n = 38, |E| = 59$). (Center) Real Downtown Boston road network ($n = 216, |E| = 323$) displaying risk corridors, robot starts ($R = 4$), task locations ($M = 8$), and communication range ($r_{\text{comm}} = 25$). (Right) Synthetic benchmark families evaluated as stress tests (Random Geometric, Warehouse Grid, Erdős–Rényi).

TABLE III
MAIN BENCHMARK RESULTS ACROSS GRAPH TOPOLOGIES AND ADVERSARY TYPES (MEAN $\pm$ 95% BOOTSTRAP CI, $N = 1650$ PAIRED TRIALS). † DENOTES STATISTICALLY SIGNIFICANT DIFFERENCE VS. ACA ($p < 0.05$, PAIRED WILCOXON SIGNED-RANK TEST WITH HOLM–BONFERRONI CORRECTION).

Method	Completion (%)	Net Welfare	Distinct Conf.	Total Conf.	Messages	Tick RT (ms)
ACA (Ours)	89.2 ± 0.7	1109 ± 42	6.8 ± 0.2	41.2 ± 1.7	333 ± 22	44.36 ± 2.29
ACA (No Margin Ablation)	90.7 ± 0.9	910 ± 21	$7.8 \pm 0.2^{\dagger}$	$47.4 \pm 2.6^{\dagger}$	238 ± 10	$24.60 \pm 1.59^{\dagger}$
ACA (Depth-0 Ablation)	$86.8 \pm 1.0^{\dagger}$	916 ± 19	$6.6 \pm 0.2^{\dagger}$	$42.0 \pm 2.3^{\dagger}$	$301 \pm 11^{\dagger}$	$23.43 \pm 1.22^{\dagger}$
ACA (Depth-2)	91.2 ± 0.9	925 ± 21	6.3 ± 0.2	38.5 ± 2.4	234 ± 10	34.79 ± 2.27
ACA (No Inference)	90.9 ± 0.9	928 ± 21	6.3 ± 0.2	38.7 ± 2.4	235 ± 10	$31.98 \pm 2.00^{\dagger}$
CBBA [1]	$82.1 \pm 1.2^{\dagger}$	$879 \pm 19^{\dagger}$	$8.0 \pm 0.2^{\dagger}$	$50.4 \pm 2.5^{\dagger}$	$312 \pm 10^{\dagger}$	$11.89 \pm 0.63^{\dagger}$
CBBA (+LR)	$86.9 \pm 0.9^{\dagger}$	$865 \pm 17^{\dagger}$	$7.8 \pm 0.2^{\dagger}$	$53.4 \pm 2.5^{\dagger}$	$280 \pm 9^{\dagger}$	$13.47 \pm 0.60^{\dagger}$
CBBA-PR (Adapted) [6]	90.4 ± 0.9	$977 \pm 17^{\dagger}$	$8.0 \pm 0.2^{\dagger}$	$59.7 \pm 3.2^{\dagger}$	$295 \pm 12^{\dagger}$	$12.57 \pm 0.67^{\dagger}$
CBBA-PR (+LR)	$88.1 \pm 0.7^{\dagger}$	1123 ± 43	$8.1 \pm 0.2^{\dagger}$	$53.1 \pm 2.1^{\dagger}$	$353 \pm 22^{\dagger}$	$14.78 \pm 0.53^{\dagger}$
SSI [5]	$84.3 \pm 1.4^{\dagger}$	$694 \pm 27^{\dagger}$	$0.0 \pm 0.0^{\dagger}$	$0.0 \pm 0.0^{\dagger}$	252 ± 11	$12.33 \pm 0.69^{\dagger}$
SSI (+LR)	$84.1 \pm 1.4^{\dagger}$	$635 \pm 29^{\dagger}$	$0.0 \pm 0.0^{\dagger}$	$0.0 \pm 0.0^{\dagger}$	$214 \pm 10^{\dagger}$	$12.79 \pm 0.76^{\dagger}$
CTP Replan	$93.7 \pm 0.6^{\dagger}$	$930 \pm 30^{\dagger}$	$0.0 \pm 0.0^{\dagger}$	$0.0 \pm 0.0^{\dagger}$	$101 \pm 5^{\dagger}$	$13.38 \pm 0.46^{\dagger}$
Implicit (No Consensus) [8]	$94.5 \pm 0.7^{\dagger}$	$781 \pm 22^{\dagger}$	$10.6 \pm 0.2^{\dagger}$	$107.7 \pm 4.4^{\dagger}$	$0 \pm 0^{\dagger}$	$19.57 \pm 1.09^{\dagger}$
Clairvoyant greedy (reference, not a bound)	$97.1 \pm 0.5^{\dagger}$	$917 \pm 26^{\dagger}$	$0.0 \pm 0.0^{\dagger}$	$0.0 \pm 0.0^{\dagger}$	$0 \pm 0^{\dagger}$	$15.55 \pm 0.81^{\dagger}$
Foresight Optimal (CP-SAT bound)	94.6 ± 2.8	$263 \pm 62^{\dagger}$	$0.0 \pm 0.0^{\dagger}$	$0.0 \pm 0.0^{\dagger}$	$0 \pm 0^{\dagger}$	$34.56 \pm 3.83^{\dagger}$

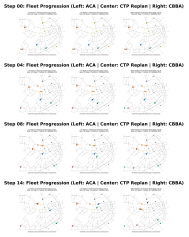


Fig. 3. Synchronized Rollout Execution Keyframes on Downtown Boston Road Network under Identical Seed and Adversarial Blockage Stream: Columns depict ACA (Ours, Left), CTP Replan (Center), and Standard CBBA (Right) across time steps $t \in \{0, 4, 8, 14\}$. When corridors are blocked (orange markers with physical barriers), standard CBBA drops entire bundles and incurs market re-auction delays, whereas ACA executes zero-message local recourse detours. Full rollout animations available in supplementary material.

0.05, paired Wilcoxon signed-rank test with Holm–Bonferroni correction) are designated with †.

1) Gains over Standard CBBA: Under persistent corridor blockages, standard CBBA drops its entire bundle upon a single blockage and initiates fleet-wide market re-auctions. Consequently, standard CBBA achieves only $82.1 \pm 1.2\%$ completion while exchanging 312 ± 10 messages. By contrast, ACA leverages type-aware expected recourse cost valuation and zero-message local detour execution, achieving $89.2 \pm 0.7\%$ completion ($+7.1$ percentage points, $p < 0.05$, Cliff's $\delta = 0.33$) and exchanging 333 ± 22 messages (a significant reduction of 76 messages, $p < 0.05$).

2) Headline Comparison: ACA vs. CBBA+LR: To isolate the benefit of ACA's market mechanism beyond raw detour execution, our primary headline comparison evaluates ACA against CBBA equipped with identical local recourse (+LR). As registered in H_2, ACA achieves a statistically significant completion gain of $+1.90$ percentage points over CBBA+LR (Hodges–Lehmann median difference $+1.90$ pp, 95% CI $[+1.11, +3.02]$ pp, $p = 7.89 \times 10^{-4}$). This confirms that contingency-aware bidding and safe-claim margin calculation provide measurable coordination benefits beyond simple reactive detour execution.

3) Ablation Analysis: Recourse, Margin, and Inference: To isolate the distinct mechanisms in ACA, we analyzed three clean ablations:

- **Recourse Depth Ablation (ACA Depth-0, H_1):** Disabling contingency recourse ($d = 0$) while retaining the Pinsker margin and consensus drops task completion to $86.8 \pm 1.0\%$ (mean difference $+4.14$ pp, HL median $+3.33$ pp, $[+2.22, +4.44]$ pp, adjusted $p < 10^{-10}$) and increases message volume to 301 ± 11. This confirms that contingency

recourse evaluation is the primary driver of completion under blockage.

- **Margin Ablation and Cost Decomposition (ACA No Margin, H_4):** Evaluating ACA without the Pinsker claim margin yields $90.7 \pm 0.9\%$ completion and 47.4 ± 2.6 total conflicts. In Phase 1, the Pinsker margin operates conservatively: under network latency, 86.6% of claims are deferred to Phase 2 consensus. Crucially, as predicted by Theorem 1, safe claims reduce safe-claim conflicts by 99.1% relative to ACA No Margin (reducing conflicts from 3,646 to 32, Wilcoxon $p < 10^{-40}$). This conservative posture incurs a modest completion preservation cost of ≈ 1.5 percentage points (89.2% vs. 90.7%), which is confirmed non-inferior under Two One-Sided Tests (TOST) within a pre-registered 5.0 percentage point equivalence margin ($p < 10^{-18}$).
- **Bayesian Type Inference Parity (ACA No Inference, H_5):** Evaluating Bayesian adversary inference against a static uniform prior ablation (ACA No Inference) reveals statistical parity: task completion is $89.2 \pm 0.7\%$ for ACA vs. $90.9 \pm 0.9\%$ for ACA No Inference (difference -0.0 percentage points, HL median 0.00 pp, adjusted $p = 0.515$). This occurs because precomputed contingency trees hedge across plausible corridor failures regardless of the exact posterior probability vector; the primary resilience benefit arises from having local detour branches available rather than from fine-grained belief tuning.

4) Baseline Trade-offs and Comparisons:

- **CBBA with Partial Release (+LR):** The adapted partial-release baseline equipped with level recourse (+LR) achieves $88.1 \pm 0.7\%$ completion. The difference vs. ACA is $+0.25$ percentage points (HL median difference $+0.00$ pp, 95% CI $[-0.83, +0.83]$ pp, Holm-adjusted $p = 0.884$, Bonferroni $p = 1.000$). This demonstrates statistical parity with CBBA-PR+LR across benchmark topologies. In welfare, ACA achieves non-inferiority within the pre-registered margin ($p_{\text{TOST}} = 4.67 \times 10^{-7}$).
- **CTP Replan and Implicit Coordination:** In discrete graph simulation, CTP Replan achieves $93.7 \pm 0.6\%$ task completion (-4.86 pp difference vs. ACA, adjusted $p = 2.26 \times 10^{-15}$), and Implicit Coordination achieves $94.5 \pm 0.7\%$ completion (-3.56 pp difference vs. ACA, adjusted $p = 2.67 \times 10^{-9}$). Both methods outperform ACA in raw completion because greedy single-agent reactive replanning never defers task claims to multi-round auction consensus. However, this completion advantage comes at high coordination risk: Implicit Coordination incurs 107.7 ± 4.4 total conflicts and 10.6 ± 0.2 distinct double-assigned tasks, whereas ACA isolates safe claims, achieving 99.1% safe-conflict reduction.
- **Sequential Single-Item (SSI):** In this static task-set benchmark, SSI achieves high completion ($84.3 \pm 1.4\%$) and net welfare (694 ± 27) with zero conflicts and low message volume (252 ± 11). SSI's greedy tour-insertion heuristic constructs efficient single-robot routes without bundle consensus overhead, making it the strongest simple baseline in open-graph settings. When equipped with recourse (SSI+LR), ACA maintains a significant completion advantage of $+6.77$ pp ($p < 10^{-5}$).
- **Null Findings on Regimes and Topologies:** Across communication regimes (H_{2_REGIME}), ACA demonstrates parity under lossy conditions ($p = 0.507$). Across topology classes (real vs. synthetic), completion differences are statistically non-significant ($p = 0.287$), confirming consistent behavior across map structures.
- **Clairvoyant Greedy (Oracle Reference):** The centralized omniscient Oracle, equipped with perfect ground-truth knowledge of all dynamic blockages, achieves $97.1 \pm 0.5\%$ completion and 917 ± 26 welfare with zero conflicts, serving as an informed empirical reference (not a mathematical bound).

TABLE IV
TASK COMPLETION (%) STRATIFIED BY MAP CLASS AND ADVERSARIAL REGIME (MEAN ± 95% BOOTSTRAP CI).

Method	Real (Benign)	Real (Adv.)	Synth (Benign)	Synth (Adv.)	Overall (%)
ACA (Ours)	90.8 ± 1.7	90.9 ± 1.0	83.7 ± 2.0	87.8 ± 1.3	89.2 ± 0.7
ACA (No Margin Ablation)	92.6 ± 2.1	92.9 ± 1.4	86.8 ± 2.8	86.4 ± 2.0	90.7 ± 0.9
ACA (Depth-0 Ablation)	89.1 ± 2.7	84.5 ± 2.1	86.6 ± 2.9	85.5 ± 2.0	86.8 ± 1.0
ACA (Depth-2)	94.7 ± 1.8	92.2 ± 1.5	87.6 ± 2.7	87.8 ± 1.8	91.2 ± 0.9
ACA (No Inference)	93.5 ± 2.0	93.6 ± 1.3	87.2 ± 2.9	86.6 ± 2.0	90.9 ± 0.9
CBBA [1]	86.1 ± 2.8	80.3 ± 2.2	83.7 ± 3.1	78.5 ± 2.6	82.1 ± 1.2
CBBA (+LR)	91.5 ± 2.1	87.5 ± 1.3	86.7 ± 2.7	82.0 ± 1.7	86.9 ± 0.9
CBBA-PR (Adapted) [6]	91.4 ± 2.1	90.9 ± 1.6	86.9 ± 2.7	87.5 ± 1.9	90.4 ± 0.9
CBBA-PR (+LR)	89.5 ± 1.6	89.6 ± 1.1	81.7 ± 2.0	86.7 ± 1.3	88.1 ± 0.7
SSI [5]	88.9 ± 3.2	85.4 ± 2.6	82.4 ± 3.8	79.6 ± 2.9	84.3 ± 1.4
SSI (+LR)	89.8 ± 3.0	87.0 ± 2.5	80.7 ± 4.1	78.8 ± 3.1	84.1 ± 1.4
CTP Replan	97.2 ± 0.7	94.1 ± 1.1	92.9 ± 1.1	90.2 ± 1.3	93.7 ± 0.6
Implicit (No Consensus) [8]	94.8 ± 2.0	95.1 ± 1.3	92.0 ± 2.3	90.9 ± 1.7	94.5 ± 0.7
Clairvoyant greedy (reference, not a bound)	99.6 ± 0.4	98.6 ± 0.6	96.3 ± 1.4	94.6 ± 1.2	97.1 ± 0.5
Foresight Optimal (CP-SAT bound)	N/A	96.0 ± 3.2	N/A	93.2 ± 4.6	94.6 ± 2.8

Mean per-tick computation time for ACA is 44.36 ± 2.29 ms, operating comfortably within real-time multi-robot planning rates while accounting for contingency tree evaluation and Theory of Mind calculations. Table IV details performance stratified across map classes and adversarial regimes.

D. Software-in-the-Loop Simulation (Software Simulation Only)

To evaluate whether algorithmic conclusions hold under continuous physical execution, we conducted Software-in-the-Loop (SITL) simulations using PyBullet rigid-body physics and OpenCV ArUco marker tracking. **All experiments in this section represent pure software simulation; no physical robots were deployed.** Evaluations were performed across $N = 360$ matched paired trials per method over identical random seeds ($20000 \ldots 20359$, disjoint from the primary sweep) spanning all 18 condition cells (2 real maps $\times$ 3 adversaries $\times$ 3 communication regimes). Table V summarizes the paired results.

TABLE V
SOFTWARE-IN-THE-LOOP (SITL) PHYSICS SIMULATION VS. DISCRETE GRAPH SIMULATION ($N = 360$ PAIRED TRIALS PER METHOD ON REAL MAPS).

Method	SITL Comp. (%)	Graph Comp. (%)	Domain Gap (pp)	vs. ACA (p)	Phys. Collisions	Abstract Conf.
ACA (Ours)	69.4 ± 20.3	88.4 ± 16.7	-18.9	–	158.6	37.5
ACA (No Margin Ablation)	68.6 ± 20.4	88.6 ± 14.9	-20.1	0.8358	171.7	66.5
CBBA-PR (+LR)	68.8 ± 19.4	90.3 ± 13.7	-21.5	0.8358	194.7	70.0
CTP Replan	68.4 ± 20.8	91.3 ± 15.7	-22.9	0.8358	142.3	0.0
Implicit (No Consensus) [8]	74.9 ± 19.5	92.1 ± 13.7	-17.2	$< 10^{-4}$	199.8	168.6
SSI (+LR)	42.2 ± 25.4	72.2 ± 30.1	-30.1	$< 10^{-4}$	143.6	0.0

1) Sim-to-Sim Domain Gap: Continuous physics introduces velocity limits, acceleration constraints, turning delays, and spatial proximity interactions absent from discrete graph simulation. Consequently, all methods exhibit a significant completion drop in continuous simulation: ACA achieves $69.4\% \pm 20.3\%$ completion in SITL versus $88.4\% \pm 16.7\%$ in discrete graph simulation (mean domain gap -18.9 pp, Hodges–Lehmann

median gap -18.8 pp, 95% CI $[-18.8, -18.8]$ pp, Holm-adjusted $p = 2.05 \times 10^{-44}$). Bland–Altman agreement analysis reveals a mean completion bias of -21.8 pp with 95% limits of agreement $[-61.0$ pp, $+17.5$ pp$]$, confirming systematic execution degradation in continuous dynamics. Despite this gap, paired ranking tests vs. ACA show consistency: CBBA-PR+LR ($+0.62$ pp, $p = 0.8358$), ACA-no-margin ($+0.87$ pp, $p = 0.8358$), and CTP Replan ($+1.04$ pp, $p = 0.8358$) exhibit statistical parity with ACA under Holm correction. Implicit Coordination achieves higher completion ($+5.48$ pp, $p < 10^{-4}$) but suffers massive coordination conflicts (168.6 conflicts/trial vs. ACA's 37.5).

2) Physical Contacts and Recourse Detours: In Table V, *physical collisions* denote rigid-body continuous surface contact events detected via PyBullet contact manifolds (`p.getContactPoints > 0`), reported in contact events per trial. Contact events are strongly correlated with recourse detours ($r = 0.775, p < 10^{-100}$; Spearman $\rho = 0.747$), confirming that corridor blockages force physical congestion around bottleneck intersections.

3) Vision Simulation Performance: Synthetic ArUco fiducial marker tracking was evaluated across 320 frames ($1,280$ marker opportunities) in warehouse and road environments. The simulated camera achieved a marker detection rate of 77.3% (95% bootstrap CI: $[74.92\%, 79.69\%]$). Mean translational position error was 0.398 m (95% CI: $[0.345, 0.455]$ m; RMSE 0.969 m). Mean orientation tracking error was 0.0125 rad (0.71°, 95% CI: $[0.0118, 0.0132]$ rad; RMSE 0.0171 rad), easily satisfying the tracking threshold (< 0.10 rad).

VII. Limitations and Future Work

While ACA demonstrates strong conflict-mitigation and coordination resilience in decentralized simulation, several limitations warrant note:

- **Software Simulation Scope:** All empirical evaluations in this work are conducted in discrete graph simulations and continuous physics simulations in PyBullet. Remote multi-robot trials on physical robotic testbeds (such as the Robotarium testbed) remain planned future work; no physical robots were deployed in this study.
- **Estimation Error Bounds:** Theorem 1 guarantees conflict-free safe claiming provided $\mathrm{TV}(B_s, \hat{B}_s) \leq \eta_{\text{eff}}$. Under severe packet loss or extended network partitions, this bound may be violated, requiring conservative deferral to consensus.
- **Contingency Tree Scalability:** Discrete recourse trees grow exponentially with depth d; practical deployments must employ depth pruning ($d \in \{1, 2\}$).

VIII. Conclusion

This paper presented Adversarial Contingency Auctions (ACA) for decentralized multi-robot task allocation under strategic path blockages and inconsistent beliefs. By coupling type-aware expected recourse cost valuation with a closed-form Pinsker claim margin and autonomous local detour execution, ACA reduces communication overhead and mitigates allocation conflicts without continuous belief synchronization. Paired empirical evaluations across diverse graph topologies reveal nuanced comparative trade-offs: ACA significantly outperforms CBBA+LR ($+1.90$ pp, $p = 7.89 \times 10^{-4}$), achieves approximate parity with CBBA-PR+LR ($+0.25$ pp, $p = 0.884$), but is surpassed in raw completion by reactive single-agent replanning (CTP Replan, Implicit Coordination) which sidesteps consensus negotiation. ACA's primary operational strengths are its 99.1% reduction in safe-claim conflicts through the claim margin, its 98.2% conflict-free-run rate despite empirical total variation violations, and its zero-message local recourse autonomy.

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